



Description

CS 258 Fall 2025 FlagMaker Student Engagement Survey

Principal Investigator: Dr. Srishti Srivastava

Department: Computer Science, Management and Information Science

Faculty Supervisor: Dr. Srishti Srivastava

Study Title: Flagmaker activity to introduce parallel and distributed computing to CS1 and CS2 students

IRBNet ID: 2255885-1

Time Requirement: under 10 minutes

Purpose: This survey asks you about the in-class activities on parallel computing as applied to the Flagmaker. We will use the results to evaluate these activities for possible use in future terms and at other institutions. The survey should take no more than 10 minutes. There is no explicit reward for completing it, but we appreciate your input and will use it to improve computing education for future students. We do not believe that your participation carries any risk to you. Your responses will be confidential. If you choose not to participate, simply do not take the survey. You may change your mind about participating at any time prior to submitting the survey. Do not take the survey if you selected "I do not consent" on the consent form provided to you at the beginning of this class.

If you have any questions about this study, you can discuss it with the principal investigator of this study, Dr. Srishti Srivastava (email: fsrishti@usi.edu).

For questions about your rights as a research participant or to discuss problems, complaints, or concerns about a research study, or to obtain information or offer input, contact the University of Southern Indiana Office of Sponsored Projects and Research, 8600 University Blvd., Evansville, IN 47712, 812-465-7000, rcr@usi.edu.

Recap of the informed consent collected during the beginning of the class on the pre-test form.

Location of the research: The location of data collection by the PI will be in one of the computer science classrooms in the Business and Engineering building at USI, using class computers or student laptops. The time commitment required from the student will account for one class session and is designed in a way that it integrated with a learning objective of the course and does not deter students from the scheduled coursework. No minors will be involved in the study. The downloaded data will be stored in a private USI Onedrive folder that only the PI can access until all the data is analyzed. The results of the survey will be analyzed using popular statistical analysis techniques such as z-tests, t-tests, sentiment analysis tools, and others.

Benefits of the research to society and/or the individual: The anticipated benefits to the participant and future students enrolled in CS258 and CS358 and eventually CS1 and CS2 students across the nation include increased knowledge in a more modern computational model including parallelism and distribution.

How may I withdraw from the study, before, during, or after taking the survey?

Before taking the survey, the students have an option to volunteer or withdraw from the survey using this informed consent. Withdrawing from the study will not impact the course outcome or performance. If students begin the survey and change their mind, they can withdraw from the study by closing the tab on their computer window, and letting the PI know about their decision. If the students want to withdraw after completing the survey, they can contact the PI to exclude their responses from the study. The Principal Investigator of this study, Dr. Srishti Srivastava, can be contacted via email at fsrishti@usi.edu.

How long data will be kept before it is destroyed?

The deidentified, coded data will be stored until the conclusion of the project on the PIs office computer and PI's USI OneDrive and then it will be destroyed. For any external members of the project, deidentified encrypted data will be provided in a shared project resource.

What will happen if I participate?

Your participation will take one class session in your current course. You will be asked to complete an online survey about your PDC experience that you gained after participating in the class activity called the Flagmaker. It is ok to answer, "No experience at all". The class session will be divided into the following: one you complete this consent form and consent "YES" to participate, you will be taken to a page that will have (i) a pre test that includes five fundamental questions on PDC topics. The test will be used to understand how much you already know about PDC fundamentals before participating in the class activity. It is ok to answer to the best of your knowledge. Any wrong answer will not impact your class grade. All answers will only be used by the PI to understand the knowledge level of participating students before the activity. The results will only be used for research purposes and will not in any way impact the final grade for the student in this class. The pre test will be followed by (ii) the Flagmaker activity, which will further be followed by (ii) a post test. The post test will have the same 5 questions that are on the pre test. The students can now answer the questions based on what they learnt about PDC from the Flagmaker activity. Further, the last part of the class session will have (iv) a survey to collect information from the students on their experience with learning the PDC concepts during the Flagmaker activity.

What are my options?

Your role as a participant in this study will include allowing de-identified data from the tests/surveys to be used in study results published to the computer science community. All information taken from the study will be coded to protect your name. No personally identifiable information will be included with your data and your identity will be fully protected. Once the data has been fully analyzed, it will be destroyed. Your decision to participate is completely voluntary, and you have the right to terminate your participation without penalty. If you do not wish to complete this survey, simply select "I do not consent" and close the form. The Principal Investigator of this study, Dr. Srishti Srivastava, can be contacted via email at fsrishti@usi.edu, if you should have any questions about this research study. The instructor of the class, Dr. Xue Han, will not know which students have chosen to participate and which have not until after the semester has ended.

How will my information be used?

Efforts will be made to keep your personal information confidential. Absolute confidentiality cannot be guaranteed. All information taken from the study will be coded to protect your name. No personally identifiable information will be included with your data

and your identity will be fully protected. Once the data has been fully analyzed, it will be destroyed. Your identity will not be shared if the study findings are published or presented. Only Dr. Srishti Srivastava will have access to the collected data. Any data that will be disseminated to other institutions or the National Science Foundation will be completely coded, anonymized, and deidentified. All collected information will be stored in a cloud drive maintained and protected under USI's policies.

What are my risks?

There are no risks to individuals participating in this survey beyond those that exist in daily life.

May I be eligible for compensation?

You may be eligible for extra credit towards your course work. Please contact your instructor or Dr. Srishti Srivastava for questions about extra credit.

Who can answer my questions about this study?

If you have questions about the study or encounter a problem with the research, contact the researcher, Dr. Srishti Srivastava at fsrishti@usi.edu.

For questions about your rights as a research participant or to discuss problems, complaints, or concerns about a research study, or to obtain information or offer input, contact the University of Southern Indiana Office of Sponsored Projects and Research, 8600 University Blvd., Evansville, IN 47712, 812-465-7000, rcr@usi.edu.

The research team members are not acting as agents of the University of Southern Indiana and do not have authority to bind the University. Any opinions, findings, or recommendations expressed in this study represent the researchers' own and do not necessarily reflect the views of the University of Southern Indiana.

To what extent do you agree or disagree with the following statements?

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
Explaining the material to my group improved my understanding of it	☐	☐	☐	☐	☐
Having the material explained to me by my group members	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
improved my understanding of it					
Group discussions during the activity contributed to my understanding of parallel computing	☐	☐	☐	☐	☐
I had fun during the activity	☐	☐	☐	☐	☐
Overall, the other members of my group made valuable contributions during the activity	☐	☐	☐	☐	☐
I would prefer to take a class that includes this group activity over one that does not	☐	☐	☐	☐	☐
I am confident in my understanding of the material presented during the activity	☐	☐	☐	☐	☐
The activity increased my understanding of parallel computing	☐	☐	☐	☐	☐
The activity stimulated my interest in parallel computing	☐	☐	☐	☐	☐
The activity increased my understanding of loops	☐	☐	☐	☐	☐
I made a valuable contribution to my group during the activity	☐	☐	☐	☐	☐
I was focused during the activity	☐	☐	☐	☐	☐
I worked hard during the activity	☐	☐	☐	☐	☐
The instructor seemed prepared for the activity	☐	☐	☐	☐	☐
The instructor put a good deal of effort into	☐	☐	☐	☐	☐

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
my learning from the activity					
The instructor's enthusiasm made me more interested in the activity	☐	☐	☐	☐	☐
The instructor was available to answer questions during the activity	☐	☐	☐	☐	☐
I like that the activity tied into the class's current programming topics	☐	☐	☐	☐	☐

What is the most interesting thing you learned from the activity?

How would you improve the activity for future classes?

Demographics

USI Email Address:

(only used to connect your responses at the beginning and end of the activity)

Please indicate your ethnicity (you can select more than one) (optional)

- ☐ American Indian or Alaska Native
 - ☐ Asian / Pacific Islander
 - ☐ Black/African American
 - ☐ Hispanic
 - ☐ Middle Eastern or North African
 - ☐ White
-

Please indicate your gender (optional)

- ☐ Male
 - ☐ Female
 - ☐ Non-binary / third gender
-

What is your major? (select all that apply)

- ☐ Computer Science
- ☐ Basic Engineering
- ☐ Chemical Engineering
- ☐ Civil Engineering
- ☐ Computer Engineering
- ☐ Engineering Technology / MET
- ☐ Electrical Engineering
- ☐ Mechanical Engineering
- ☐ Math
- ☐ Physics
- ☐ Other

